Logarithmic algebraic geometry: logification and the ghost sheaf

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October 28, 2026

1 Limits and colimits of commutative monoids

As a preparation for what follows, we briefly discuss limits and colimits of monoids.

Lemma 1.1. *The category* CMon *admits all limits.*

Proof. The limit of underlying sets can be endowed with a monoid structure. This is the limit in CMon. □

Colimits also exist, but their construction is more involved.

Fact 1.2. *The category* CMon *admits all colimits.*

As a particularly important non-trivial example, we give the following construction of the pushout in a special case.

Lemma 1.3. *Let $\theta_1 \colon Q \to P_1$ and $\theta_2 \colon Q \to P_2$ be two homomorphisms of monoids. On $P_1 \times P_2$, we consider a relation given by $(p_1, p_2) \sim (r_1, r_2)$ if and only if there are $q, q' \in Q$ with $p_1 + \theta_1(q) = r_1 + \theta_1(q')$ and $p_2 + \theta_2(q') = r_2 + \theta_2(q)$. This is an equivalence relation, and the monoid structure on $P_1 \times P_2$ descends to $(P_1 \times P_2)/\sim$. If P_2 is a group, then the square*

$$\begin{array}{ccc} Q & \xrightarrow{\theta_1} & P_1 \\ \downarrow{\scriptstyle \theta_2} & & \downarrow \\ P_2 & \longrightarrow & (P_1 \times P_2)/\sim \end{array}$$

is a pushout in the category of commutative monoids.

Remark 1.4. I leave the proof of this as an exercise. If P_1 or Q is a group, the same formula holds. If none of the three monoids is a group, the pushout is computed by a more complicated formula. ◊

Similarly, we have limits and colimits of sheaves of commutative monoids.

Fact 1.5. *Let X be a topological space. Then,* $\mathrm{Shv}(X, \mathrm{CMon})$ *admits all limits and colimits.*

Example 1.6. Let X be a topological space, and let $\theta_1 \colon \mathcal{Q} \to \mathcal{P}_1$ and $\theta_2 \colon \mathcal{Q} \to \mathcal{P}_2$ be homomorphisms of sheaves of monoids. Assume that $\mathcal{P}_2$ is a sheaf of Abelian groups. Then, we compute the pushout in $\mathrm{Shv}(X, \mathrm{CMon})$ by applying the above formula over every open subset, and sheafifying afterwards. ◊

2 Logification

Any pre-log structure can be turned into a log structure, in a universal way. Applying this to *constant* pre-log structures is a second avenue to the construction of log structures, besides compactifying log structures. Neither approach is strictly more general than the other, but they have a substantial overlap.

Proposition 2.1. *Let $\beta\colon \mathcal{P} \to \mathcal{O}_X$ be a pre-log structure. Then, there is a factorization $\mathcal{P} \to \mathcal{P}^{\log} \xrightarrow{\alpha} \mathcal{O}_X$ such that $\alpha\colon \mathcal{P}^{\log} \to \mathcal{O}_X$ is a log structure, and such that any other factorization $\mathcal{P} \to \mathcal{M} \to \mathcal{O}_X$ with $\gamma\colon \mathcal{M} \to \mathcal{O}_X$ a log structure admits precisely one monoid homomorphism $\delta\colon \mathcal{P}^{\log} \to \mathcal{M}$ such that*

$$\begin{array}{ccccc} \mathcal{P} & \longrightarrow & \mathcal{P}^{\log} & \xrightarrow{\alpha} & \mathcal{O}_X \\ \| & & \downarrow{\scriptstyle \delta} & & \| \\ \mathcal{P} & \longrightarrow & \mathcal{M} & \xrightarrow{\gamma} & \mathcal{O}_X \end{array}$$

commutes.

Proof. The sheaf $\mathcal{P}^{\log}$ is constructed as the pushout

$$\begin{array}{ccc} \beta^{-1}(\mathcal{O}_X^\times) & \longrightarrow & \mathcal{P} \\ \downarrow & & \downarrow \\ \mathcal{O}_X^\times & \longrightarrow & \mathcal{P}^{\log} \end{array}$$

in the category of sheaves of monoids on X. The map $\alpha_X\colon \mathcal{P}^{\log} \to \mathcal{O}_X$ is constructed via the universal property. By applying the formula to compute the pushout—which is valid because $\mathcal{O}_X^\times$ is a sheaf of Abelian groups at the stalks—one can see that $\alpha_X^{-1}(\mathcal{O}_X^\times) \to \mathcal{O}_X^\times$ is an isomorphism. Uniqueness and existence of $\delta\colon \mathcal{P}^{\log} \to \mathcal{M}$ follows from combining the universal property of the pushout with the log condition $\gamma^{-1}(\mathcal{O}_X^\times) \cong \mathcal{O}_X^\times$. □

Definition 2.2. Let $(X, \mathcal{P})$ be a pre-log scheme. Then, the log scheme $(X, \mathcal{P}^{\log})$ is the *logification* of $(X, \mathcal{P})$. We have a morphism of pre-log schemes $(X, \mathcal{P}^{\log}) \to (X, \mathcal{P})$. ♦

Example 2.3. (1) Let $S_0 = \operatorname{Spec}\mathbb{C}$, and let $\beta\colon \underline{\mathbb{N}}_{S_0} \to \mathcal{O}_{S_0}$ be given by $\beta(0) = 1$ and $\beta(1) = 0$. Then, $\alpha\colon (\underline{\mathbb{N}}_{S_0})^{\log} \to \mathcal{O}_{S_0}$ is a log structure on $S_0 = \operatorname{Spec}\mathbb{C}$. It is not a divisorial log structure for any closed subset $D \subset S_0$.

(2) Let $P = P_\sigma$ be a toric monoid. Then, we have a pre-log structure $\beta\colon \underline{P}_{A_P} \to \mathcal{O}_{A_P}$. Moreover, we have the divisorial log structure $\alpha\colon \mathcal{M}_{A_P} \to \mathcal{O}_{A_P}$, and a homomorphism $\phi\colon \underline{P}_{A_P} \to \mathcal{M}_{A_P}$. Thus, we obtain a homomorphism $\delta\colon (\underline{P}_{A_P})^{\log} \to \mathcal{M}_{A_P}$. It is in fact an isomorphism, but this is not obvious.

(3) Sometimes a log structure $\mathcal{M}_X$ on a scheme X can be written *locally* as $(\underline{P}_U)^{\log} \to \mathcal{O}_U$, for open subsets $U \subseteq X$. For an example, let $(X, \mathcal{M}_X) = (\mathbb{P}^1 | \{0, \infty\})$. Then, we have open subset $U_1 = \mathbb{P}^1 \setminus \{\infty\} \cong \operatorname{Spec}\mathbb{C}[x]$ and $U_2 = \mathbb{P}^1 \setminus \{0\} \cong \operatorname{Spec}\mathbb{C}[y]$. On U_1, we have a pre-log structure $\beta_1\colon \underline{\mathbb{N}}_{U_1} \to \mathcal{O}_{U_1}$ given by $\beta_1(1) = x$. Then, $(\underline{\mathbb{N}}_{U_1})^{\log} \to \mathcal{M}_X|_{U_1}$ is an isomorphism. Similarly, we have an isomorphism $(\underline{\mathbb{N}}_{U_2})^{\log} \cong \mathcal{M}_X|_{U_2}$.

(4) Not every divisorial log structure on an affine scheme X can be written as $(\underline{P}_X)^{\log}$. For an example, let $X = \operatorname{Spec}\mathbb{C}[t,x,y,u]/(xy-tu)$, let $D = \{t=0\} \cong \operatorname{Spec}\mathbb{C}[x,y,u]/(xy)$, and let $(X,\mathcal{M}_X) = (X|D)$. Then, there is *no* $\beta\colon \underline{P}_X \to \mathcal{M}_X$ such that $(\underline{P}_X)^{\log} \to \mathcal{M}_X$ would be an isomorphism. (This is not obvious.) ◊

Exercise 2.4. Show that $(\underline{P}_{A_P})^{\log} = \mathcal{M}_{(A_P|D_P)}$ for $P = \mathbb{N}$.

3 Spectra of pre-log rings

We introduce the algebraic analog of (pre-)log schemes.

Definition 3.1. A *pre-log ring* is a commutative ring R together with a homomorphism of commutative monoids $\beta\colon P \to (R,\cdot)$. A *log ring* is a pre-log ring such that $\beta^{-1}(R^\times) \to R^\times$ is an isomorphism. A *homomorphism* of pre-log rings $\varphi\colon (\gamma\colon Q \to A) \to (\beta\colon P \to R)$ consists of a ring homomorphism $\varphi_\sharp\colon A \to R$ and a monoid homomorphism $\varphi_\flat\colon Q \to P$ such that $\beta \circ \varphi_\flat = \varphi_\sharp \circ \gamma$. We write $\mathrm{CRing}^{\mathrm{plog}}$ for the category of pre-log rings, and $\mathrm{CRing}^{\log}$ for its full subcategory of log rings. ♦

When $(X,\mathcal{M}_X)$ is a log scheme, we obtain a log ring $\Gamma(X,\mathcal{M}_X) \to \Gamma(X,\mathcal{O}_X)$. Conversely, let $(\beta\colon P \to R)$ be a pre-log ring. On the scheme $X = \operatorname{Spec} R$, we obtain a pre-log structure $\underline{P}_X \to \mathcal{O}_X$ induced by $\beta\colon P \to R$. Then, its logification $(\underline{P}_X)^{\log} \to \mathcal{O}_X$ is a log structure. We denote the resulting log scheme by $\operatorname{Spec}(\beta\colon P \to R)$.

Example 3.2. Let $P = P_\sigma$ be a toric monoid. Then, we have $A_P^\dagger = \operatorname{Spec}(P \to \mathbb{C}[P])$. ◊

Example 3.3. Let Q be a sharp toric monoid. Then, $\beta\colon Q \to \mathbb{C}$ given by $\beta(0) = 1$ and $\beta(q) = 0$ for $q \neq 0$ is a pre-log ring. We say that $S_0^\dagger = \operatorname{Spec}(Q \to \mathbb{C})$ is a *log point*. One may show that $\Gamma(S_0,\mathcal{M}_{S_0}) = Q \oplus \mathbb{C}^\times$.

If $Q \neq 0$, there is no closed subset $D \subset S_0$ with $S_0^\dagger \cong (S_0|D)$. Thus, not every logification of a constant pre-log structure is a compactifying log structure.

For $Q = \mathbb{N}$, we say that $\operatorname{Spec}(\mathbb{N} \to \mathbb{C})$ is the *standard log point*. For $Q = 0$, we say that $\operatorname{Spec}(0 \to \mathbb{C})$ is the *trivial log point*. Indeed, the log structure is trivial in this case. ◊

Fact 3.4. *The following statements hold:*

(1) *Global sections form a functor* $\Gamma\colon \mathrm{Sch}^{\log} \to (\mathrm{CRing}^{\log})^{\mathrm{op}} \subseteq (\mathrm{CRing}^{\mathrm{plog}})^{\mathrm{op}}$.

(2) *The spectrum forms a functor* $\operatorname{Spec}\colon (\mathrm{CRing}^{\mathrm{plog}})^{\mathrm{op}} \to \mathrm{Sch}^{\log}$.

(3) Γ *is a left adjoint to* Spec. *In other words, we have a canonical isomorphism*

$$\mathrm{Mor}((X,\mathcal{M}_X),\operatorname{Spec}(P \to R)) \cong \mathrm{Hom}(P \to R, \Gamma(X,\mathcal{M}_X) \to \Gamma(X,\mathcal{O}_X)).$$

Example 3.5. Let $(P \to R) = (\mathbb{N} \to \mathbb{C})$. As a pre-log ring, this has no non-trivial automorphism over $(0 \to \mathbb{C})$. The logification of this pre-log ring is $(\mathbb{N} \oplus \mathbb{C}^\times \to \mathbb{C})$. Now, there are non-trivial automorphisms given by $\varphi_\flat(n,\lambda) = (n, \lambda \cdot \mu^n)$ for $\mu \in \mathbb{C}^\times$. They are precisely the automorphisms of $S_0^\dagger = \operatorname{Spec}(\mathbb{N} \to \mathbb{C})$ over $\operatorname{Spec}(0 \to \mathbb{C})$. In particular, $\operatorname{Spec}\colon (\mathrm{CRing}^{\mathrm{plog}})^{\mathrm{op}} \to \mathrm{Sch}^{\log}$ is not fully faithful. ◊

Warning 3.6. In the case of schemes, the functor $\operatorname{Spec}\colon (\mathrm{CRing})^{\mathrm{op}} \to \mathrm{Sch}$ is fully faithful because $R \to \Gamma \circ \operatorname{Spec}(R)$ is an isomorphism. Evidently, this fails for a pre-log ring $(P \to R)$ since $\Gamma(X,\mathcal{M}_X) \to \Gamma(X,\mathcal{O}_X)$ is always a log ring. However, even if $(P \to R)$ is a log ring, $(P \to R) \to \Gamma \circ \operatorname{Spec}(P \to R)$ is not necessarily an isomorphism. Moreover, $\operatorname{Spec}\colon (\mathrm{CRing}^{\log})^{\mathrm{op}} \to \mathrm{Sch}^{\log}$ is indeed not fully faithful. ◊

Example 3.7. Let $R = \mathbb{C}[t]$, $P = \mathbb{N} \oplus \mathbb{C}^\times$, and $\beta\colon P \to R$ be given by $\beta(n, \lambda) = \lambda \cdot t^n \cdot (t-1)^n$. Let $(X, \mathcal{M}_X) = \mathrm{Spec}(P \to R)$. Then, $\mathcal{M}_X$ can be identified with the divisorial log structure given by $D = \{0, 1\}$. In particular, we have $t \in \Gamma(X, \mathcal{M}_X)$, but $t \in \Gamma(X, \mathcal{O}_X)$ is not in the image of $\beta\colon P \to R$. ◊

4 The ghost sheaf

Let P be a monoid, and let $F \subseteq P$ be a face. Then, we can form the *quotient* P/F by setting $p_1 \sim p_2$ if there are $f_1, f_2 \in F$ with $p_1 + f_1 = p_2 + f_2$. For the face $F = P^\times$, we write $P/P^\times =: \overline{P}$. This is called the *sharpening* of P.

Example 4.1. (1) Let $P = \mathbb{N}^{\oplus 2}$ and $F = \mathbb{N} \cdot (0, 1)$. Then, we have $P/F \cong \mathbb{N}$ via $(m, n) \mapsto m$.

(2) Let $P = \mathbb{N} \oplus \mathbb{Z}$. Then, we have $\overline{P} \cong \mathbb{N}$ via $(m, n) \mapsto m$.

(3) In general, we have $P/F \cong P_F/P_F^\times$. ◊

We now introduce one of the main tools to understand the geometry of a log structure.

Definition 4.2. Let $(X, \mathcal{M}_X)$ be a log scheme. Then, the *ghost sheaf* is

$$\overline{\mathcal{M}}_X \coloneqq \mathcal{M}_X/\mathcal{M}_X^\times.$$

More precisely, it is the sheafification of the presheaf

$$W \mapsto \Gamma(W, \mathcal{M}_X)/\Gamma(W, \mathcal{M}_X)^\times.$$

Exercise 4.3. Let $(X, \mathcal{M}_X)$ be a log scheme, and let $x \in X$ be a point. Then, we have $\overline{\mathcal{M}}_{X,x} \coloneqq (\overline{\mathcal{M}}_X)_x = \mathcal{M}_{X,x}/\mathcal{M}_{X,x}^\times$.

Example 4.4. Let $(X, \mathcal{O}_X^\times)$ be a trivial log scheme. Then, we have $\overline{\mathcal{M}}_X = 0$. Conversely, if $(X, \mathcal{M}_X)$ is a log scheme with $\overline{\mathcal{M}}_X = \{0\}$, then $\mathcal{M}_X = \mathcal{O}_X^\times$. ◊

Proposition 4.5. *Let $P = P_\sigma$ be a toric monoid, and let $(X, \mathcal{M}_X) = A_P^\dagger$. Let $x \in V_{P,F}^\circ \subset X$. Then, the composition $\underline{P}_{X,x} \to \overline{\mathcal{M}}_{X,x}$ can be identified with $P \to P/F$.*

Proof. Let $\beta\colon \underline{P}_X \to \mathcal{O}_X$ be the pre-log structure, and recall that $\mathcal{M}_X = (\underline{P}_X)^{\log}$. Thus, we have a pushout square

$$\begin{array}{ccc} \beta^{-1}(\mathcal{O}_{X,x}^\times) & \longrightarrow & P \\ \downarrow & & \downarrow \\ \mathcal{O}_{X,x}^\times & \longrightarrow & \mathcal{M}_{X,x} \end{array}$$

in CMon. Since $F = F_{\mathfrak{p}} = \{p \in P \mid \mathrm{z}^p \notin \mathfrak{p}\}$ for the prime ideal $\mathfrak{p}$ corresponding to x, we have $\beta^{-1}(\mathcal{O}_{X,x}^\times) = F$. It is now a straightforward computation in the explicit formula for the pushout that $P \to \overline{\mathcal{M}}_{X,x}$ can be identified with $P \to P/F$. □

Example 4.6. (1) Let $P = \mathbb{N}$. For $x = 0$, we have $\overline{\mathcal{M}}_{X,x} = \mathbb{N}/0 = \mathbb{N}$, and for $x \in \mathbb{A}^1 \backslash \{0\}$, we have $\overline{\mathcal{M}}_{X,x} = \mathbb{N}/\mathbb{N} = 0$.

(2) Let $P = P_2$. For $x = 0$, we have $\overline{\mathcal{M}}_{X,x} = P_2/0 = P_2$. For $F = \mathbb{N} \cdot (0,1)$ and $x \in V^\circ_{P,F}$, we have $\overline{\mathcal{M}}_{X,x} = P_2/F \cong \mathbb{N}$. For $G = \mathbb{N} \cdot (-1,2)$ and $x \in V^\circ_{P,G}$, we have $\overline{\mathcal{M}}_{X,x} = P/G \cong \mathbb{N}$. For $x \in A^\circ_P$, we have $\overline{\mathcal{M}}_{X,x} = P/P = 0$.

(3) Let $\sigma = \mathbb{R}_{\geq 0} \cdot \{(1,0,0),(1,1,0),(1,0,1),(1,1,1)\}$. Then, we have

$$\sigma^\vee = \mathbb{R}_{\geq 0} \cdot \{(0,1,0),(1,-1,0),(1,0,-1),(0,0,1)\}$$

and $\mathbb{C}[P_\sigma] \cong \mathbb{C}[t,x,y,u]/(xy - tu)$. The monoid P_σ has one face of rank zero, four faces of rank one, four faces of rank two, and one face of rank three. In $0 \in V^\circ_{P,P}$, we have $\overline{\mathcal{M}}_{X,0} = P_\sigma$, and in all other points, $\overline{\mathcal{M}}_{X,x}$ is a quotient thereof. ◊

Lemma 4.7. *Let $f\colon (X,\mathcal{M}_X) \to (Y,\mathcal{M}_Y)$ be a morphism of log schemes. Then, we have an induced morphism $f^+\colon \overline{\mathcal{M}}_Y \to f_*\overline{\mathcal{M}}_X$.*

Proof. Let $m \in \Gamma(W,\mathcal{M}_Y)$, and let $u \in \Gamma(W,\mathcal{M}_Y^\times)$. Then, we have $f^\flat(u) \in \Gamma(W, f_*\mathcal{M}_X)^\times$, and thus $f^\flat(m) = f^\flat(m+u) \in \Gamma(W, f_*\overline{\mathcal{M}}_X)$. □

Example 4.8. Let $P = \mathbb{N}^{\oplus 3}$, let $Q = \mathbb{N}$, and let $\theta\colon Q \to P$ be given by $\theta(1) = (1,1,1)$. Then, we obtain a homomorphism of pre-log rings $(\mathbb{N} \to \mathbb{C}[\mathbb{N}]) \to (P \to \mathbb{C}[P])$, and hence a morphism of log schemes $\mu\colon A^\dagger_P \to A^\dagger_{\mathbb{N}}$. On the level of underlying schemes, we have $A_{\mathbb{N}} = \operatorname{Spec}\mathbb{C}[t]$ and $A_P = \operatorname{Spec}\mathbb{C}[x,y,z]$, and the map is given by $\mu^\sharp(t) = xyz$. In particular, we have $D_P = \{xyz = 0\} = \mu^{-1}(0)$. For a point $x \in A^\circ_P$, the map $f^+\colon \overline{\mathcal{M}}_{A_{\mathbb{N}},\mu(x)} \to \overline{\mathcal{M}}_{A_P,x}$ is just $0 \to 0$. For a point $x \in D_P$, it is given by $\mathbb{N} \to P \to P/F$ for the face $F \subseteq P$ with $x \in V^\circ_{P,F}$. ◊

5 Strict morphisms

Let $f\colon X \to Y$ be a morphism of schemes, and let $\alpha_Y\colon \mathcal{M}_Y \to \mathcal{O}_Y$ be a log structure on Y. Then, we obtain a pre-log structure $f^{-1}\mathcal{M}_Y \to f^{-1}\mathcal{O}_X \to \mathcal{O}_Y$. We set

$$f^*_{\log}\mathcal{M}_Y \coloneqq (f^{-1}\mathcal{M}_Y)^{\log},$$

the logification.

Definition 5.1. A morphism of log schemes $f\colon (X,\mathcal{M}_X) \to (Y,\mathcal{M}_Y)$ is *strict* if

$$f^*_{\log}\mathcal{M}_Y \to \mathcal{M}_X$$

is an isomorphism. ⧫

Fact 5.2. *The following statements hold:*

(1) *Compositions of strict morphisms are strict.*

(2) *If $f\colon (X,\mathcal{M}_X) \to (Y,\mathcal{M}_Y)$ is strict, then $f^+\colon f^{-1}\overline{\mathcal{M}}_Y \to \overline{\mathcal{M}}_X$ is an isomorphism.*

(3) *If $f^+\colon f^{-1}\overline{\mathcal{M}}_Y \to \overline{\mathcal{M}}_X$ is an isomorphism, and if $\mathcal{M}_X$ is integral, then f is strict.*

(4) *Let $\varphi\colon (Q \to A) \to (P \to R)$ be a homomorphism of pre-log rings with $\varphi_\flat\colon Q \to P$ an isomorphism. Then, $\operatorname{Spec}(P \to R) \to \operatorname{Spec}(Q \to A)$ is strict.*

Example 5.3. We have a homomorphism of pre-log rings $(Q \to \mathbb{C}[Q]) \to (Q \to \mathbb{C})$, given by $\mathrm{z}^0 \mapsto 1$ and $\mathrm{z}^q \mapsto 0$ for $q \neq 0$. It induces a strict morphism of log schemes $S^\dagger_0 \to A^\dagger_Q$. To see this, compute the ghost sheaves. ◊

6 Fiber products

Recall that the category of sheaves of monoids on a topological space X has pushouts.

Proposition 6.1. *Let $f\colon X^\dagger \to Z^\dagger$ and $g\colon Y^\dagger \to Z^\dagger$ be morphisms of log schemes. Then, a fiber product $W^\dagger = X^\dagger \times_{Z^\dagger} Y^\dagger$ exists in $\mathrm{Sch}^{\log}$.*

Proof. On the level of underlying schemes, we set $W \coloneqq X \times_Z Y$. Let $p\colon W \to X$ and $q\colon W \to Y$ be the two projections. Let $r = f \circ p = g \circ q$. Then, we have two morphisms of log structures $r^*_{\log}\mathcal{M}_Z \to p^*_{\log}\mathcal{M}_X$ and $r^*_{\log}\mathcal{M}_Z \to q^*_{\log}\mathcal{M}_Y$. Let $\mathcal{P}_W$ be the pushout in the category of sheaves of monoids on W. Then, we set $\mathcal{M}_W = \mathcal{P}_W^{\log}$, the logification of $\mathcal{P}_W \to \mathcal{O}_W$. □

Lemma 6.2. *Let $f\colon (X, \mathcal{M}_X) \to (Y, \mathcal{M}_Y)$ be a morphism of log schemes. Then, f is strict if and only if*

$$\begin{array}{ccc} (X, \mathcal{M}_X) & \xrightarrow{f} & (Y, \mathcal{M}_Y) \\ \downarrow & & \downarrow \\ (X, \mathcal{O}_X^\times) & \longrightarrow & (Y, \mathcal{O}_Y^\times) \end{array}$$

is Cartesian.

Proof. Note that $f^*_{\log}\mathcal{M}_Y$ is the logification of $\mathcal{O}_X^\times \oplus_{f^*_{\log}\mathcal{O}_Y^\times} f^*_{\log}\mathcal{M}_Y$, and hence coincides with the log structure of the fiber product. Thus, $f^*_{\log}\mathcal{M}_Y \to \mathcal{M}_X$ is an isomorphism if and only if the diagram is Cartesian. □

Corollary 6.3. *Let $f\colon (X, \mathcal{M}_X) \to (Y, \mathcal{M}_Y)$ be a morphism of log schemes. Then, f is strict if and only if it has the following universal property: Let $h\colon (Z, \mathcal{M}_Z) \to (Y, \mathcal{M}_Y)$ be a morphism of log schemes, and let $g\colon Z \to X$ be a morphism of schemes such that $f \circ g = h$ as morphisms of schemes. Then, there is a unique enhancement to a morphism of log schemes $h\colon (Z, \mathcal{M}_Z) \to (X, \mathcal{M}_X)$ with $f \circ g = h$ as morphisms of log schemes.*

In the category of pre-log rings $\mathrm{CRing}^{\mathrm{plog}}$, pushouts can be computed as

$$\begin{array}{ccc} (Q \to A) & \longrightarrow & (P_1 \to R_1) \\ \downarrow & & \downarrow \\ (P_2 \to R_2) & \longrightarrow & (P_1 \oplus_Q P_2 \to R_1 \otimes_A R_2). \end{array}$$

It then follows from the adjunction that

$$\mathrm{Spec}(P_1 \to R_1) \times_{\mathrm{Spec}(Q \to A)} \mathrm{Spec}(P_2 \to R_2) \cong \mathrm{Spec}(P_1 \oplus_Q P_2 \to R_1 \otimes_A R_2).$$

Example 6.4. Let $Q = \mathbb{N}$ and $P = \mathbb{N}^{\oplus 3}$, giving rise to $\mu\colon A_P^\dagger \to A_{\mathbb{N}}^\dagger$. Furthermore, we have the strict morphism $S_0^\dagger \to A_{\mathbb{N}}^\dagger$ induced by $(\mathbb{N} \to \mathbb{C}[t]) \to (\mathbb{N} \to \mathbb{C})$. Then, the fiber product is $M^\dagger = \mathrm{Spec}(\beta\colon \mathbb{N}^{\oplus 3} \to \mathbb{C}[x, y, z]/(xyz))$ with $\beta(1, 0, 0) = x$, $\beta(0, 1, 0) = y$, and $\beta(0, 0, 1) = z$. ◊